

# Executive summary and recommendations

This project analyzed global sugarcane production data covering 103 countries to understand patterns in output, land use, and efficiency. Brazil emerged as the world's leading producer, accounting for approximately **39%** of total global sugarcane production, while India contributed nearly **18%**, and China about **6%**. Together, the top three countries represented over **63%** of global output. Asia as a continent led in production volume, followed closely by South America. Data cleaning involved converting numeric fields from string formats, handling missing values (notably, **1%** of acreage and yield entries were missing), and renaming columns for clarity. Visual analysis highlighted a strong positive correlation between acreage and production, but yield per hectare varied significantly across nations. For example, Brazil achieved high production due to large acreage (**over 10 million hectares**), while some smaller countries demonstrated superior yields despite limited land. Boxplots and bar charts underscored disparities in productivity among continents, and scatter plots revealed outliers with exceptional efficiency. The study suggests that adopting best practices from high-yield countries could help underperforming regions enhance production. Overall, the findings underscore the critical role of a few nations in meeting global sugarcane demand and the need for targeted agricultural investments to boost yields sustainably.

- **Brazil alone produces ~39% of global sugarcane**, making it the undisputed leader.
- **India contributes ~18%**, ranking second in total output.
- **China accounts for ~6%**, placing third globally.
- **63% of global production** comes from the top 3 countries.
- **Asia is the largest producing continent**, followed by South America.
- **1% of records** had missing data for acreage and yield, which were addressed during cleaning.
- A **strong positive correlation (>80%)** exists between acreage size and total production.
- Yield per hectare varies by **over 50% between the lowest and highest performing countries**, indicating significant efficiency gaps.
- Smaller nations can achieve **exceptionally high yields**, demonstrating that land size alone is not the sole determinant of output.